

# Filtered Fresh Air Supply Units WITH EC MOTOR

## 700DD-EC SERIES

FEATURING EC MOTORS  
- UP TO 70% EFFICIENCY



ENERGY SAVINGS OF UP TO 70% OVER  
REGULAR DUTY PSC MOTORS

High efficiency direct drive motor  
for energy efficient low maintenance  
applications

### FEATURES

- Grey, weather resistant baked powder coat finish.
- Latched top cover provides convenient service access to 4 filters.
- EC Motor offers true variable speed operation and up to 70% energy savings over PSC motors.

### PERFORMANCE SPECIFICATIONS

| MODEL      | MOTOR |                |         | TYPICAL MAXIMUM CFM @ SP |      |      |      |             |      |      |      |             |      |      |      |             |      |      |      |             |        |      |      |             |
|------------|-------|----------------|---------|--------------------------|------|------|------|-------------|------|------|------|-------------|------|------|------|-------------|------|------|------|-------------|--------|------|------|-------------|
|            | HP    | Variable Speed | Max RPM | Speed                    | 1/4" |      |      |             | 1/2" |      |      |             | 3/4" |      |      |             | 1"   |      |      |             | 1 1/4" |      |      |             |
|            |       |                |         |                          | CFM  | BHP  | FEI  | FEPact (KW) | CFM  | BHP  | FEI  | FEPact (KW) | CFM  | BHP  | FEI  | FEPact (KW) | CFM  | BHP  | FEI  | FEPact (KW) | CFM    | BHP  | FEI  | FEPact (KW) |
| 709DD050EC | 1/2   | Y*             | 1200    | High                     | 1850 | -    | -    | -           | 1650 | -    | -    | -           | 1425 | -    | -    | -           | 1250 | -    | -    | -           | 500    | -    | -    | -           |
|            |       |                |         | Low                      | 600  | -    | -    | -           | 500  | -    | -    | -           | 500  | -    | -    | -           | 500  | -    | -    | -           | 450    | -    | -    | -           |
| 709DD100EC | 1     | Y*             | 1200    | High                     | 2400 | 0.99 | 0.84 | 0.88        | 2275 | 0.99 | 0.93 | 0.88        | 2087 | 0.97 | 0.98 | 0.87        | 1400 | 0.59 | 1.16 | 0.55        | 472    | 0.28 | 1.16 | 0.28        |
|            |       |                |         | Low                      | 600  | 0.06 | 2.00 | 0.09        | 500  | 0.10 | 1.66 | 0.13        | 500  | 0.16 | 1.42 | 0.18        | 500  | 0.22 | 1.28 | 0.23        | 400    | 0.26 | 1.13 | 0.27        |
| 710DD050EC | 1/2   | Y*             | 1100    | High                     | 1900 | -    | -    | -           | 1400 | -    | -    | -           | 1050 | -    | -    | -           | 750  | -    | -    | -           | 500    | -    | -    | -           |
|            |       |                |         | Low                      | 600  | -    | -    | -           | 500  | -    | -    | -           | 500  | -    | -    | -           | 500  | -    | -    | -           | 400    | -    | -    | -           |
| 710DD100EC | 1     | Y*             | 1100    | High                     | 2600 | 0.97 | 0.83 | 0.87        | 2250 | 0.98 | 0.83 | 0.87        | 1900 | 0.99 | 0.80 | 0.88        | 1600 | 0.77 | 0.99 | 0.70        | 700    | 0.59 | 0.77 | 0.55        |
|            |       |                |         | Low                      | 600  | 0.10 | 1.44 | 0.09        | 500  | 0.16 | 1.17 | 0.18        | 500  | 0.26 | 0.95 | 0.27        | 500  | 0.37 | 0.95 | 0.36        | 400    | 0.44 | 1.13 | 0.27        |
| 712DD050EC | 1/2   | Y*             | 1000    | High                     | 2650 | -    | -    | -           | 2300 | -    | -    | -           | 1900 | -    | -    | -           | 1425 | -    | -    | -           | 1100   | -    | -    | -           |
|            |       |                |         | Low                      | 800  | -    | -    | -           | 600  | -    | -    | -           | 600  | -    | -    | -           | 600  | -    | -    | -           | 500    | -    | -    | -           |
| 712DD100EC | 1     | Y*             | 1000    | High                     | 3575 | 1.00 | 1.09 | 0.89        | 3250 | 0.98 | 1.17 | 0.87        | 2950 | 0.98 | 1.22 | 0.87        | 2675 | 0.99 | 1.25 | 0.88        | 2000   | 0.80 | 1.30 | 0.72        |
|            |       |                |         | Low                      | 800  | 0.08 | 1.96 | 0.11        | 600  | 0.11 | 1.70 | 0.13        | 600  | 0.18 | 1.42 | 0.20        | 600  | 0.25 | 1.27 | 0.26        | 600    | 0.33 | 1.15 | 0.33        |

\* Minimum motor speed 425 RPM (EC Models)

EC motors can be controlled using the built in potentiometer on motor or by using a 0 to 10 V controller, or by a remote potentiometer.  
REVIEW FAN CURVES BEFORE ADJUSTING RPM TO ENSURE LOAD DOES NOT EXCEED MOTOR HP.

### DIMENSIONS

| MODEL | A       | B       | C       | D        | E         | F       | G        | H        | J       | FILTERS  |          | *INSIDE CABINET | WEIGHT (lbs) without motor |
|-------|---------|---------|---------|----------|-----------|---------|----------|----------|---------|----------|----------|-----------------|----------------------------|
|       |         |         |         |          |           |         |          |          |         | SIZE     | QUANTITY |                 |                            |
| 709DD | 24 3/8" | 32 3/8" | 24 3/8" | 10 1/4"  | 11 13/16" | 6 9/32" | 7 1/16"  | 7 1/16"  | 10 1/4" | 10"x 20" | 4        | 24"x 24"        | 84                         |
| 710DD | 28 3/8" | 38 1/2" | 28 1/4" | 11 3/8"  | 13 1/8"   | 7 5/8"  | 8 1/2"   | 8 1/2"   | 12 1/4" | 12"x 24" | 4        | 28"x 28"        | 108                        |
| 712DD | 32 3/8" | 42 5/8" | 31 3/8" | 13 7/16" | 15 5/8"   | 8 3/8"  | 9 15/32" | 9 15/32" | 16 1/2" | 16"x 25" | 4        | 32"x 32"        | 150                        |

\* Curb size to be smaller than inside cabinet to allow for flashing and roofing.

